

1 Drone Job Market Analysis

1.1 Executive Summary

This research validates the core hypothesis behind Cuesta College's drone program structure: mapping/photogrammetry represents the 1 commercial drone application, and thermal imaging applications (spanning both public safety and infrastructure inspection) represent the 2 application. The program's focus on these sectors is well-founded.

A key finding is that infrastructure inspection, particularly utility work for PG&E, represents the single largest local employment opportunity. PG&E deploys 80+ drone pilots daily across California and logged over 300,000 inspections in 2024, with drone docks operational through Paso Robles. This argues strongly for combining the originally conceived "public safety" course with infrastructure inspection training, unified by thermal imaging as the core technical skill.

The program serves two student populations: existing professionals (surveyors, engineers, first responders) seeking to add drone capabilities, and early career students entering the industry. The three-course Phase One structure serves both effectively. Phase Two expansion into GIS analysis leverages instructor expertise and positions graduates for higher-level data analyst roles.

Recommended Phase One Structure:

1. AERO 100: Basic UAV Operator
2. Drone Mapping and Photogrammetry
3. Thermal Imaging and Applied Inspection (combined public safety and infrastructure)

Recommended Phase Two Expansion:

4. GIS Analysis and Drone Data Processing
5. Agricultural Drone Operations (multispectral, with potential spray pathway)

1.2 Market Overview

1.2.1 Global and National Scale

The commercial drone market was valued at approximately \$30 billion in 2024 and is projected to reach \$54-94 billion by 2030-2033, depending on the estimate. North America accounts for roughly 40% of the global market.

The top commercial drone applications, ranked by current market activity:

1. **Mapping and Surveying** - Consistently ranked as the 1 commercial drone application
2. **Inspection** - Second largest application, with infrastructure inspection projected to grow 50% in the next five years
3. **Photography and Filming** - Media and entertainment accounts for 21.4% of drone services revenue
4. **Agriculture** - Fastest-growing segment at 9.28% CAGR, with some projections suggesting 80% of future drone usage could come from this sector

1.2.2 California Job Market

Current job listings in California show:

- 116 drone pilot positions on Indeed
- 408 drone-related jobs on LinkedIn
- Geographic hotspots: Bay Area, Los Angeles, and increasingly the Central Coast

Key hiring sectors in California:

- Energy and utility inspection (driven largely by wildfire prevention requirements)

- Construction and surveying
- Agriculture (particularly in the Central Valley)
- Real estate and media production
- Drone delivery (emerging, concentrated in Bay Area)

1.3 Sector Analysis

1.3.1 Mapping and Surveying

Market Position: The dominant commercial drone application across multiple industries including construction, mining, public administration, and waste management. The only industry where mapping does not play a significant role is insurance.

Salary Range: \$60,000-150,000 annually, with higher pay for those with GIS backgrounds, surveying credentials, or LiDAR experience. Construction-focused positions typically fall in the \$90,000-110,000 range.

Local Relevance: High. San Luis Obispo County has active construction and civil engineering sectors. Multiple firms including Walsh Engineering, Lea & Braze Engineering, and JW Design & Construction operate locally. The wine industry drives substantial construction and land development activity.

Employment Model: Mix of full-time positions at surveying/engineering firms and contract work. Larger construction companies increasingly hire in-house pilots.

Assessment: Strong fit for the program. Color photogrammetry with RGB cameras is the entry point for most mapping work, and the equipment requirements align with current program holdings.

1.3.2 Public Safety

Market Position: Growing rapidly with Drone as First Responder (DFR) programs spreading across California. The 2-19kg drone segment is the fastest-growing hardware category specifically due to public safety adoption.

Key California Programs:

- Chula Vista Police Department: First DFR program in the nation (2018)
- Fremont Police/Fire: First joint public safety DFR program in the country (2022-2024)
- Los Angeles Fire Department: Pioneered wildfire reconnaissance drones (2015)
- Corona Fire Department, Santa Monica Police, San Diego Fire-Rescue all operate active programs

Local Relevance: Moderate to high. SLO County fire departments face wildfire risks similar to other California agencies. Search and rescue operations in the coastal and mountain terrain present ongoing needs. However, most public safety drone positions are filled by existing sworn personnel who receive drone training, rather than external hires.

Employment Model: Most public safety drone pilots are existing firefighters, police officers, or SAR volunteers who add drone certification to their skillset. Dedicated civilian drone pilot positions in public safety are rare. This course may be most valuable as supplemental training for students pursuing public safety careers, or for existing first responders seeking professional development.

Assessment: Valuable offering, but employment pathway is indirect. Students may need to pursue public safety careers first, then leverage drone skills internally. Consider partnerships with local fire and police agencies for both curriculum input and potential hiring pipelines.

1.3.3 Infrastructure and Utility Inspection

Market Position: Second-largest commercial drone application. Critical for energy, telecommunications, and transportation sectors. Projected to grow 50% over the next five years.

Local Relevance: Very high. PG&E operates Diablo Canyon Power Plant and extensive transmission infrastructure in SLO County. PG&E logged over 300,000 drone inspections in 2024 and typically deploys around 80 drone pilots daily across California. They have drone docks operational from Redding through Paso Robles.

Employment Model: Mix of PG&E employees and contractors. The utility uses both in-house pilots and contracted inspection services. Entry-level positions pay \$22-48/hour for inspection work.

Salary Range: \$50,000-100,000 for general inspection work; higher for specialized assets like wind turbines or transmission lines.

Assessment: This represents a significant local opportunity that the current program structure may underserve. Consider whether the public safety course could incorporate or be renamed to include infrastructure inspection, or whether inspection deserves its own track. Thermal imaging skills transfer directly between public safety and utility inspection applications.

1.3.3.1 Deep Dive: PG&E and Utility Inspection

PG&E represents the single largest potential employer for drone pilots in the Central Coast region. Their drone program, operational since 2015, has grown into one of the most extensive utility drone operations in the country.

Scale of Operations:

- 180-drone fleet across California
- 80+ pilots deployed daily during peak inspection periods
- 300,000+ pole and tower inspections completed in 2024
- Drone docks (autonomous charging stations) operational from Redding to Paso Robles
- 12 dedicated pilots for autonomous operations, plus 3 for remote control room operations

Equipment Standards: PG&E uses DJI Matrice series drones and Skydio autonomous platforms equipped with high-resolution cameras, thermal sensors, and in some cases LiDAR. The Matrice 4T that Cuesta already owns aligns with PG&E's equipment standards.

Hiring Requirements:

- FAA Part 107 certification (mandatory)
- Understanding of GO 95 and GO 165 (California utility safety regulations)
- Flight experience with commercial UAV platforms
- Utility inspection experience preferred but not always required
- Physically demanding work: pilots must hike to remote sites, often in difficult terrain
- Garmin inReach satellite communication devices provided for remote operations

Career Pathway: PG&E hires both employees and contractors. The program was built up by Kellen Kirk, a former U.S. Navy aircraft mechanic and commercial pilot. Entry-level positions focus on visual inspections; advancement involves BVLOS operations and specialized payload work.

BVLOS Operations: PG&E holds FAA waivers for Beyond Visual Line of Sight operations, which are increasingly important for utility inspection. Pilots in these roles must comply with strict telemetry logging requirements and anti-collision protocols.

1.3.3.2 Other Infrastructure Inspection Opportunities

Beyond utilities, infrastructure inspection encompasses several sub-sectors:

Solar Farm Inspection:

- Thermal imaging to identify hotspots, broken panels, and connection failures
- California has extensive solar installations requiring regular inspection
- Often combined with utility work at the same employers

Wind Turbine Inspection:

- Close-range blade inspection for wear, cracks, and surface damage
- Nacelle and tower infrastructure assessment
- Higher-paying specialty due to technical demands
- Limited local opportunity (wind farms concentrated elsewhere in California)

Cell Tower Inspection:

- Telecommunications companies use drones to inspect towers without sending climbers
- Electromagnetic interference is a technical consideration
- Growing sector as 5G infrastructure expands

Bridge and Transportation Infrastructure:

- CalTrans and county public works departments increasingly use drones
- Concrete beam inspection, deck assessment, and structural documentation
- Potential partnership opportunity with SLO County

1.3.3.3 Training and Certification for Infrastructure Inspection

Beyond Part 107, infrastructure inspection careers benefit from:

Thermography Certification:

- Critical skill for utility, solar, and building inspection work
- Level 1 certification teaches theory, operation, and data interpretation
- Can significantly increase earning potential (one pilot reported going from \$30k to \$200k after thermography specialization)
- Training costs: \$1,500-\$5,000

Utility-Specific Knowledge:

- GO 95 (California overhead electric line construction rules)
- GO 165 (inspection requirements for electric distribution/transmission facilities)
- Understanding of electrical infrastructure components and failure modes

Safety Protocols:

- Working near energized lines requires specific safety training
- Electromagnetic interference awareness
- Most operators maintain at least 100 feet clearance from power lines

1.3.3.4 Salary Progression in Infrastructure Inspection

- Entry-level powerline inspection: \$50,000-\$65,000 annually
- Mid-level (3-5 years experience): \$65,000-\$85,000
- Senior inspector or program manager: \$90,000-\$120,000+
- BVLOS specialists and those with LiDAR/specialized payload experience command premium rates

Thermography specialization can dramatically increase earning potential regardless of experience level.

1.3.4 Agriculture

Market Position: Fastest-growing application segment. California's Central Valley is a major adoption center. Primary applications include crop monitoring (multispectral imaging), precision spraying, and field mapping. The global agricultural drone market is projected to exceed \$10 billion by 2030, with some analysts suggesting 80% of future commercial drone usage could come from agriculture.

Local Relevance: Moderate for the Central Coast specifically, but high for California overall. SLO County has significant wine grape production and some row crop agriculture. The Salinas Valley (just

north) is a major agricultural center where commercial drone spraying has recently launched. The Central Valley represents the largest concentration of agricultural drone work in California.

Employment Model: Mix of positions at agricultural technology companies, farming operations, and contract spraying services. The work is often seasonal, with peak demand during growing season.

Salary Range: \$60,000-90,000 annually for full-time positions; \$40,000-80,000 for seasonal or entry-level work. Freelance spraying operators can earn significantly more during peak season.

Assessment: The Phase Two agricultural course makes strategic sense, but requires significant equipment investment and additional certifications beyond Part 107. This is a more complex pathway than mapping or inspection.

1.3.4.1 Multispectral Imaging and Crop Monitoring

Multispectral drones capture data beyond visible light to assess plant health, identify stress, and guide precision agriculture decisions.

Applications:

- NDVI (Normalized Difference Vegetation Index) mapping to assess crop health
- Early detection of pest infestations, disease, and irrigation problems
- Variable rate prescription maps for fertilizer and chemical application
- Yield estimation and harvest planning
- Research applications at universities and agricultural experiment stations

Equipment Requirements:

- Multispectral sensor (e.g., MicaSense, DJI Mavic 3 Multispectral)
- RTK-capable drone for georeferenced data
- Processing software capable of generating vegetation indices

Equipment Costs:

- DJI Mavic 3 Multispectral: approximately \$6,500
- MicaSense RedEdge-P or Altum: \$5,500-\$12,000
- Higher-end fixed-wing systems with multispectral: \$15,000-\$30,000

Employment Opportunities:

- Precision agriculture consultants
- Ag technology companies (crop analytics, farm management software)
- University research programs
- Large farming operations with in-house agronomy staff
- Vineyard management companies

Revenue Model (Contract/Freelance):

- Monthly subscriptions: \$1,000-\$3,000 per farm
- Hourly rates: \$50-\$150/hour for data collection
- Often bundled with analysis and recommendation services

Local Vineyard Opportunity: Wine grape operations on the Central Coast increasingly use drone technology for vineyard management. Clos de la Tech in the Santa Cruz Mountains has been an early adopter. AgVision CA offers AI-powered drone scouting specifically for winegrowers. This represents a potential niche for Central Coast graduates, though the market is smaller than Central Valley row crops.

1.3.4.2 Spray Operations

Agricultural spraying represents the highest-revenue segment of agricultural drone work, but also requires the most extensive certification and equipment investment.

Applications:

- Herbicide application
- Fungicide treatment
- Insecticide spraying
- Fertilizer application
- Cover crop seeding

Certification Requirements (Stacked):

1. **FAA Part 107** - Basic commercial drone certification
2. **FAA Part 137** - Agricultural Aircraft Operator Certificate
 - Same certification required for manned crop dusters
 - Covers regulations specific to aerial chemical application
 - Focuses on safety and environmental considerations
 - Application approval takes 2-4 months
3. **44807 Exemption** - Required for drones over 55 lbs
 - Most commercial spray drones (DJI Agras T40, T50, Hylio AG272) exceed this weight
 - Must be included in Part 137 exemption request
4. **California QAC/QAL License** - State pesticide applicator certification
 - Issued by California Department of Pesticide Regulation
 - Requires passing exams on laws, regulations, and pest control principles
 - Applicants must be at least 18 years old
5. **County Registration** - Annual registration with each county of operation
 - Required for legal pesticide application in California

Equipment:

- DJI Agras T40/T50: \$20,000-\$30,000 (plus batteries)
- Hylio AG272: \$25,000+
- Annual maintenance and nozzle replacement
- Chemical handling equipment and PPE
- Support vehicle for field operations

Physical Demands: Spray pilots must lift 50+ lbs regularly (chemical containers, batteries), work outside for extended periods, and often operate in remote locations. Work is seasonal and weather-dependent.

Salary and Revenue:

- Entry-level employed positions: approximately \$45,000/year plus bonuses for certifications
- Experienced operators: \$60,000-\$100,000
- Contract spraying rates: \$10-\$30 per acre depending on crop and location
- Peak season potential (freelance with own equipment): \$8,000/day spraying 400 acres at \$20/acre
- California vineyard specialists report \$150/hour for mapping plus \$25/acre for spraying

Companies Hiring in California:

- Davey Resource Group (herbicide spray drone pilots, Los Angeles area)
- nuWay Ag (Central Valley operations)
- Various farm management companies and agricultural cooperatives
- Direct employment by large farming operations

Central Coast Opportunity: The Salinas Valley has recently seen commercial drone spraying operations launch. Vineyard spraying represents a potential niche, though the terrain (steep hillsides) creates operational challenges and opportunities.

1.3.4.3 Program Implications for Agricultural Drone Training

If Cuesta pursues agricultural drone training, the program faces several decision points:

Option A: Multispectral Only

- Lower equipment investment (\$6,500 for Mavic 3 Multispectral)
- No additional FAA certifications beyond Part 107
- Prepares students for crop monitoring and precision ag consulting
- Can be combined with general mapping curriculum
- Does not prepare students for spray operations

Option B: Spray Operations

- Significant equipment investment (\$25,000+ for spray drone)
- Requires Part 137 certification curriculum
- Requires coordination with California DPR for applicator training
- Prepares students for highest-paying agricultural drone work
- More complex liability and safety considerations

Option C: Both

- Comprehensive agricultural drone program
- Highest equipment and curriculum development costs
- Creates multiple career pathways for graduates

Equipment Consideration: The program's existing Matrice 4T includes thermal imaging but not multispectral. Multispectral sensors capture specific wavelength bands (red edge, near-infrared) that thermal cameras do not. A dedicated multispectral drone would be required for crop health assessment training.

Certification Pathway: Part 137 certification could potentially be offered as a post-program continuing education option rather than embedded in the core curriculum, allowing students to complete the foundational program and add spray certification later if they pursue that career path.

1.3.5 Media, Real Estate, and Photography

Market Position: Media and entertainment accounts for 21.4% of drone services revenue, making it the third-largest application area. Real estate has been an early adopter.

Local Relevance: Present but limited. The Central Coast has real estate activity and some film/media production, but this sector is more concentrated in Los Angeles and the Bay Area.

Employment Model: Heavily freelance and contract-based. Most real estate drone work is done by independent operators or small studios like Studio 101 West Photography in SLO.

Assessment: Not recommended as a dedicated course focus. Students will gain applicable photography and videography skills through the introductory and mapping courses. The freelance nature of this sector makes it less suitable for an employment-focused program.

1.3.6 Insurance and Claims Inspection

Market Position: The global drone insurance market was valued at \$1.1 billion in 2023 and is projected to more than double by 2032. Insurance companies increasingly use drones for roof inspections, property assessments, and catastrophe response. Carriers using drone solutions can resolve up to 1.5x more claims per day.

Local Relevance: Moderate. California experiences wildfires, storms, and earthquake damage requiring property inspections. The work is seasonal and follows weather patterns, with higher demand after major weather events.

Employment Model: Mix of in-house adjusters with drone skills and independent contractors. Most carriers partner with certified pilots for inspection work. Important distinction: without an adjuster's license, drone pilots capture imagery but do not interpret damage or assess claims.

Salary Range: \$48,000-\$160,000 for insurance adjuster drone pilots. Per-inspection rates of \$100-\$300 for residential, \$500+ for commercial properties.

Curriculum Alignment: Skills from the Thermal/Inspection course transfer directly. Roof inspection techniques, thermal imaging for moisture detection, and professional documentation are core competencies for insurance work.

Assessment: Good supplementary employment pathway for graduates of the thermal/inspection course. Students interested in this sector should consider pursuing adjuster licensing separately, as this opens higher-paying assessment roles rather than imagery-only work.

1.3.7 Mining and Quarry Operations

Market Position: Volumetric surveying for stockpile measurement and pit/dump topography updates represents steady demand in mining and aggregate industries. Drones capture in 15 minutes what takes surveyors hours on foot, with better accuracy.

Local Relevance: Limited for SLO County specifically. Major mining and aggregate operations are concentrated elsewhere in California. However, graduates with these skills have mobility to pursue opportunities statewide.

Employment Model: Mix of in-house positions at mining companies and contract work through surveying firms. Positions require 2-3+ years of commercial drone experience, often preferring mining or heavy construction backgrounds.

Salary Range: Competitive base salaries of \$90,000-\$130,000 for experienced volumetric survey specialists. Entry-level positions less common; this sector typically seeks experienced operators.

Technical Requirements: Strong photogrammetry skills, ability to produce and interpret volumetric reports (cut/fill, inventory), experience with DEMs and contour maps. GIS software proficiency (ArcGIS, QGIS) highly valued.

Assessment: Excellent career advancement opportunity for mapping course graduates who gain experience and seek higher-paying specialization. Not an entry-level pathway, but volumetric analysis skills are directly taught in the mapping curriculum.

1.3.8 Environmental Monitoring and Wildlife Research

Market Position: The drone inspection and monitoring market generated over \$16 billion in 2024, with environmental monitoring as a leading growth segment. Research applications include wildlife population counts, habitat assessment, and ecosystem monitoring.

Local Relevance: Moderate to high. Central Coast ecosystems, coastal monitoring, and conservation efforts create research opportunities. Morro Bay National Estuary Program collaboration demonstrates local environmental application. Cal Poly research programs are potential employers.

Employment Model: Primarily research technician and field positions at conservation organizations, universities, and government agencies. Less corporate employment structure than other sectors.

Salary Range: Research technician positions typically \$40,000-\$60,000. Senior research positions and coordinators can reach \$70,000-\$90,000. Academic positions follow university pay scales.

Technical Requirements: Mapping skills, thermal imaging for wildlife detection, GIS analysis capabilities. Increasingly, AI-powered species identification and automated population counting are emerging requirements.

Assessment: Natural fit for students with environmental science interests. The NEP collaboration provides direct experience in environmental drone applications. Could serve as informal specialization track within existing courses rather than dedicated curriculum.

1.4 Local Employment Landscape

1.4.1 San Luis Obispo County Drone-Related Employers

Major Drone/UAS Companies:

- **Edge Autonomy** (SLO): Leader in unmanned and autonomous technology, products used in 78 countries. Hires UAS Test Pilots.
- **Zone 5 Technologies** (SLO): UAS development for government applications.
- **Inspired Flight** (SLO): Drone manufacturing.
- **WhiteFox Defense Technologies**: Counter-drone and airspace management.

Potential Drone-Using Employers:

- **PG&E/Diablo Canyon**: Major utility with extensive drone inspection operations.
- **Cal Poly**: Research applications, potential partnership opportunities.
- **Walsh Engineering, Lea & Braze Engineering**: Civil engineering and surveying firms.
- **JW Design & Construction, Rarig Construction**: Commercial construction.
- **Wine industry**: Vineyard mapping and monitoring potential.
- **County of San Luis Obispo**: Public works, emergency services.

1.4.2 Employment vs. Contracting Reality

A 2018 study found 62% of drone services providers are self-employed. However, full-time positions are becoming more common as the industry matures. Companies like PG&E, construction firms, and surveying companies increasingly hire in-house pilots rather than relying solely on contractors.

For graduates seeking employment rather than entrepreneurship:

- Target surveying/engineering firms, construction companies, and utilities
- Expect entry-level positions to pay \$50,000-70,000 with benefits
- Specialized skills (thermal imaging, data processing) command higher salaries
- Contract networks like FlyGuys and Zeitview offer work but not traditional employment

1.5 Target Student Populations

The program serves two distinct student populations with different needs and prior knowledge:

1.5.1 Existing Professionals Seeking Skill Augmentation

This group includes surveyors, environmental consultants, engineers, police officers, firefighters, home inspectors, and other professionals who want to add drone capabilities to their existing expertise. They typically have:

- Domain knowledge in their field (a surveyor understands what deliverables clients need)
- Professional networks and potential employers already identified
- Limited time availability (often working while taking courses)
- Clear understanding of how drones fit into their work

Program implications: These students benefit from courses that focus on drone operations and data collection rather than extensive background on industry context. They may prefer evening or weekend scheduling. The intro course provides Part 107 preparation and flight proficiency; advanced courses give them the specific technical skills (mapping workflows, thermal interpretation) to integrate drones into their practice.

1.5.2 Early Career Students

This group includes recent high school graduates and career changers entering the drone industry. They typically have:

- Limited or no prior professional experience
- More schedule flexibility
- Need for both technical skills AND industry context
- Interest in understanding multiple career pathways before specializing

Program implications: These students benefit from exposure to multiple applications before choosing a specialization. They need more context about industries, employers, and career pathways. Partnerships with local employers for internships or job placement become especially valuable for this population.

1.5.3 Serving Both Populations

The three-course structure accommodates both groups effectively:

1. **AERO 100 (Intro):** Essential for both populations. Professionals need Part 107; early career students need foundational skills.
2. **Mapping Course:** Valuable for surveyors, engineers, and consultants adding drone skills. Also provides early career students with the most broadly applicable commercial skill.
3. **Thermal/Inspection Course:** Attracts existing first responders, utility workers, and home inspectors. Creates employment pathways for early career students into utility inspection.

The key is ensuring course marketing and scheduling accommodate working professionals while providing sufficient industry context for those new to the field.

1.6 Recommendations

1.6.1 Proposed Course Structure: Combining Thermal Inspection and Public Safety

The research strongly supports merging the originally conceived “public safety” course with infrastructure inspection training. The unifying element is thermal imaging, which serves both sectors.

1.6.1.1 Rationale for Combination

Shared Technical Foundation:

- Both public safety and infrastructure inspection rely heavily on thermal imaging
- The Matrice 4T already owned by the program serves both applications
- Flight operations near structures and in complex environments are common to both
- Data interpretation skills (identifying heat signatures, anomalies) transfer directly

Complementary Career Pathways:

- Public safety drone positions typically go to existing sworn personnel
- Infrastructure inspection offers direct civilian employment opportunities
- A combined course serves first responders seeking professional development AND early career students seeking utility/inspection jobs
- Students gain flexibility to pursue either path based on their circumstances

Market Alignment:

- PG&E represents the largest local employer for drone pilots (80+ daily, 300,000+ annual inspections)
- Public safety agencies increasingly require thermal drone capabilities
- Solar farm inspection, building envelope assessment, and utility work all use identical skills

1.6.1.2 Proposed Course: “Thermal Imaging and Applied Inspection”**Core Competencies:**

- Infrared theory and heat transfer fundamentals
- Thermal camera operation, calibration, and image optimization
- Thermal data interpretation and anomaly identification
- Flight operations near structures, towers, and hazardous environments
- Report generation and professional documentation
- Safety protocols for energized equipment and emergency scenes

Application Modules:*Module A: Utility and Infrastructure Inspection*

- Power line and pole inspection techniques
- Solar panel thermal assessment
- Building envelope and roof inspection
- Cell tower and telecommunications infrastructure
- California GO 95/GO 165 regulatory overview

Module B: Public Safety Operations

- Search and rescue thermal techniques
- Fire scene assessment and hotspot identification
- Law enforcement tactical support applications
- Night operations and low-light environments
- Incident command integration

Equipment Requirements: The existing Matrice 4T with thermal payload is well-suited for this curriculum. No additional major equipment purchases required.

Certification Pathway: Students completing this course would be prepared for Level 1 thermography certification from organizations like Infrasppection Institute or ITC. This certification significantly increases earning potential across all thermal imaging applications.

Partnership Opportunities:

- PG&E for curriculum input and potential hiring pipeline
- Local fire departments (CAL FIRE, SLO County Fire) for public safety module input
- SLO County for infrastructure inspection case studies
- Solar installation companies for panel inspection training sites

1.6.1.3 Phase Two: GIS Analysis and Advanced Data Processing

The natural second phase for program expansion focuses on the analysis side of drone data, leveraging instructor expertise in GIS.

Rationale:

- Complements the data collection focus of mapping and inspection courses
- GIS skills significantly increase graduate employability and salary potential
- Positions graduates for roles beyond “drone pilot” into data analyst/specialist positions
- Differentiates program from basic flight training programs

Potential Curriculum Elements:

- Photogrammetry processing workflows (Metashape, Pix4D, DroneDeploy)
- GIS fundamentals and integration with drone data
- Orthomosaic production and quality assessment
- Digital elevation models and volumetric analysis
- Change detection and temporal analysis
- Data management and deliverable standards

Target Students: This course would particularly appeal to existing GIS professionals adding drone skills, surveyors seeking to bring processing in-house, and graduates of the mapping course who want to move beyond field operations into data analysis roles.

1.6.2 Updated Program Structure

Based on this research, the recommended program structure is:

Phase One (Current Planning):

1. AERO 100: Basic UAV Operator - Part 107 preparation, flight proficiency, industry overview
2. Drone Mapping and Photogrammetry - RGB orthophotos, RTK positioning, professional workflows
3. Thermal Imaging and Applied Inspection - Combined public safety and infrastructure inspection

Phase Two (Future Expansion):

4. GIS Analysis and Drone Data Processing - Post-processing, analysis, and advanced deliverables
5. Agricultural Drone Operations - Multispectral imaging, with potential spray certification pathway

1.6.3 Partnership Priorities

Based on employment opportunity and program relevance:

1. **PG&E** - Largest local drone employer, uses same equipment, clear hiring pipeline potential
2. **Local Fire/Police** - Public safety module input, continuing education market
3. **NEP (Morro Bay)** - Already developing as mapping course partner
4. **Cal Poly** - Research collaboration, potential student pipeline
5. **SLO County Public Works** - Infrastructure inspection case studies

1.6.4 Equipment Recommendations

Current Holdings (Adequate for Phase One):

- DJI Matrice 4T (thermal) - Supports inspection/public safety course
- DJI Matrice 4E (mapping) - Supports mapping course
- Training fleet (Mini 3, Air 3) - Supports intro course
- Emlid RTK system - Supports mapping course

Phase Two Additions:

- DJI Mavic 3 Multispectral (\$6,500) - Required for agricultural course
- Processing workstations - Required for GIS analysis course
- Spray drone (if pursuing spray certification) - \$25,000+ investment, evaluate based on demand

1.6.5 Sectors Appropriately Excluded

- **Media/Real Estate Photography:** Too freelance-oriented for an employment-focused program
- **Drone Delivery:** Emerging but concentrated in specific companies and regions
- **Defense/Military:** Requires security clearances; local companies (Edge Autonomy, Zone 5) may hire but this is a specialized pathway
- **LiDAR Surveying:** Equipment costs make this impractical for a community college program

1.7 Sources

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1.7.4 Utility and Infrastructure Inspection

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- The Drone Life. “The Ultimate Beginner’s Guide to Drone Power Line Inspections.” <https://thedronelifej.com/power-line-inspections-guide/>
- FlyGuys. “How Insurance Companies Are Using Drones to Transform Claims and Inspections.” <https://flyguys.com/how-insurance-companies-are-using-drones-to-transform-claims-and-inspections/>
- EagleView. “Drone Inspections for Property Insurance Claims.” <https://www.eagleview.com/product/eagleview-assess-insurance/>

1.7.5 Agricultural Drone Operations

- Fair Lifts. “Agricultural Drone Pilot Licensing: Your Pathway to a Thriving Career.” <https://www.fairlifts.com/uav/agricultural-drones/agricultural-drone-pilot-licensing-your-pathway-to-a-thriving-career-in-agrochemical-application/>
- Flying Acres LLC. “Careers.” <https://flyingacresllc.com/careers/>

1.7.6 Training and Certification Programs

- Infrasppection Institute. “Level 1 sUAS Thermography Course.” <https://infrasppection.com/product/level-1-suas-thermography-drone-distance-learning>
- DARTdrones. “Aerial Drone Inspections Workshop.” <https://www.dartdrones.com/aerial-drone-inspections/>
- The Regional Training Center. “Public Safety Drone Training.” <https://thertc.org/public-safety-drone-training/>
- ABJ Academy. “Drone Thermography Level 1 Certification.” <https://abjacademy.global/drone-training-courses/thermography-level-1-certification/>
- Drone U. “Intro to Drone Thermography.” <https://www.thedroneu.com/courses/intro-to-drone-thermography/>

1.7.7 Community College Programs

- Citrus College. “Noncredit Drone Certificate Program.” <https://www.citruscollege.edu/ce/dronetech/Pages/default.aspx>
- Harford Community College. “Drone Technology.” <https://www.harford.edu/academics/workforce-career-programs/programs/drone-technology.php>
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- Michigan Technological University. “Advanced Photogrammetry and Mapping with UAS Certificate.” <https://www.mtu.edu/gradschool/programs/certificates/photogrammetry-mapping/>

- University of Victoria. “Micro-certificate in Mapping with Drones.” <https://continuingstudies.uvic.ca/science-and-the-environment/programs/mapping-with-drones>

1.7.8 Mapping and Photogrammetry Training

- UAV Coach. “Drone Mapping Essentials Online Course.” <https://uavcoach.com/online-drone-mapping-course/>
- Iprosurv. “3 Day Drone Mapping - Surveying and Photogrammetry Course.” <https://iprosvr.com/3-day-drone-mapping-surveying-and-photogrammetry-course/>
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